Centenary Celebrated Sharnbasveshwar Vidya Vardhaka Sangha's



ಶರಣಬಸವ Sharnbasva ವಿಶ್ವವಿದ್ಯಾಲಯ University

ಕಲಬುರಗಿ–585 103
ಕರ್ನಾಟಕ–ಭಾರತ
ಸ್ಥಾಪನೆ : 2017

Kalaburagi-585 103
Karnataka-India
Estd.: 2017



| Faculty of: Engineering and Technology | Department: CSE/CSD<br>Program: B.Tech | AY: 2026-27 |
|---|---|---|

## INTERNAL ASSESSMENT (I.A.) TEST: I

| Course Title: Operating Systems | Course Code: 25CS304 |
|---|---|
| Date: 04/11/2026 | Time: 2.00 pm to 3.30 pm |
| Semester and Section: III Sem | Max. Marks:30 |

**Note: Answer any three full questions selecting at least one question from each module**

| Q. No. | | QUESTIONS | Marks | CO's | RBT Levels |
|---|---|---|---|---|---|
| | | **MODULE-I** | | | |
| 1 | a) | Identify the different types of system calls and relate each type to appropriate operating-system operations. | 06 | CO1 | L3 |
| | b) | Interview the abstract view of a computer system with a neat diagram and show how its major components interact with the operating system. | 06 | CO1 | L3 |
| | | **OR** | | | |
| 2 | a) | Utilize the mechanisms of inter-process communication and compare shared-memory and message-passing approaches. | 06 | CO1 | L3 |
| | b) | Identify the process concept and explain how process states and process operations are related during process execution | 06 | CO1 | L3 |
| | | **MODULE-II** | | | |
| 3 | a) | Compare the many-to-one, one-to-one, and many-to-many multithreading models with respect to their execution characteristics. | 06 | CO2 | L4 |
| | b) | Consider the following snapshot of a system<br><br>**Process / Arrival Time / Burst time**<br>P0 / 0 / 3<br>P1 / 2 / 6<br>P2 / 4 / 4<br>P3 / 6 / 5<br>P4 / 8 / 2<br><br>Consider the following set of processes with CPU burst time(in ms). Compute the waiting time and average turnaround time forthe above process using FCFS, SJF non preemptive scheduling algorithm. | 06 | CO2 | L3 |
| | | **OR** | | | |
| 4 | a) | Examine the benefits and challenges of multithreaded programming in multi-core systems with suitable examples. | 06 | CO2 | L4 |

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| | | | | | |
|---|---|---|---|---|---|
| | b) | Demonstrate the handling of threading issues such as fork(), exec(), signal handling, thread cancellation, and thread-local storage using suitable examples. | 06 | CO2 | L3 |
| | | **MODULE-III** | | | |
| 5 | a) | Explain the three requirements that a solution to the critical-section problem must satisfy. | 06 | CO3 | L2 |
| | | **OR** | | | |
| 6 | a) | Outline and explain the structure of the writer process and reader process. | 06 | CO3 | L2 |

**Marks Distribution of CO's and RBT Level wise**

| CO's | L1 | L2 | L3 | L4 | Total Marks | % of Marks Allotted |
|---|---|---|---|---|---|---|
| CO1 | - | - | 24 | - | 24 | 40% |
| CO2 | - | - | 12 | 12 | 24 | 40% |
| CO3 | - | 12 | | - | 12 | 20% |
| Total RBT Marks | - | 12 | 36 | 12 | 60 | 100% |
| % of RBT Levels | - | 20% | 60% | 20%- | 100% | 100% |